# ACTL4001
# Actuarial Theory & Practice A
# Group Project
# Term 1 2026
## Galaxy General Insurance Company

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**Executive Summary**

This report presents recommendations to Galaxy General's Management Team in response to the request for proposals issued by Cosmic Quarry Mining Corporation (CQMC), who are expanding mining operations to the Helionis Cluster, Bayesia System, and Oryn Delta. Following a thorough actuarial analysis of the historical claims, prospective exposure, and potential challenges, the team recommends that four insurance products are offered, covering each of the four requested hazard areas: Equipment Failure, Cargo Loss, Workers' Compensation, and Business Interruption.

## 1. Product Rationale

It is recommended that all four insurance products be offered across all three of CQMC's solar systems. While the nature and severity of risks differ between the solar systems, each of the three present exposure across all four hazard areas. Thus, excluding any product from any system would leave CQMC with significant uninsured gaps in its ERM framework. As such, offering these insurance products to CQMC is expected to be a profitable business venture for Galaxy General. However, it is to be noted that each solar system should place priority on different hazard areas according to their distinct operational and environmental characteristics.

### 1.1 Helionis Cluster

Equipment and Cargo Loss should be the priority coverage areas in this system. Helionis Cluster's two major asteroid clusters present significant risk with their erratic drift patterns, frequent micro-collisions, and shifting debris clouds. The periodic need to shield and re-position relay satellites perpetuates such risk, which can be mitigated through Business Interruptions insurance coverage, supporting its inclusion.

### 1.2 Bayesia System

The priority coverages for the Bayesia System should be Workers' Compensation and Equipment Failure. Radiation spikes, combined with the high gravity and thin magnetosphere environment, creates elevated health and safety risks and accelerated degradation of mining equipment. Business Interruption is also material, as radiation events have the potential to force unplanned operational shutdowns, for indefinite periods of time.

### 1.3 Oryn Delta

Oryn Delta presents the least predictable risk profile of the three systems. The expansion into the system's asymmetric asteroid ring makes Cargo Loss a critical priority as transit risk through this zone is high and growing. The dwarf star's potential for unpredictable flares, combined with inconsistent long-range communications supports the need for Business Interruption coverage. Workers' Compensation is also elevated given the risky nature of the expansion, while Equipment Failure risk is present but relatively more manageable.

## 2. Product Design

The team has designed four key products for Galaxy General Insurance that encompass the four hazard areas. In an endeavour of creating suitable products, Exploratory Data Analysis (EDA) was conducted on historical data, which helped in identifying the key risks and characteristics of the underlying solar systems. These insights were incorporated into the benefits structures, coverage triggers and exclusions of each product.

To assist with adaptability, each of the four hazard areas were split in response to their respective EDA analyses (*see Appendix A*). Equipment's EDA, supplemented with an average

risk index rating system, enabled the hazard to be categorised as either 'High Risk' (consisting of Quantum Bores, Regl Aggregators, and Ion Pulverizers) or 'Low Risk' (consisting of Graviton Extractors, FexStram Carriers, and Flux Riders). Cargo Loss was split based on inherent value per kilogram, calculated by dividing cargo value by weight and then averaging the results for each cargo type (*Appendix Figure 4E*). Gold and Platinum were found to be more expensive, thus classified as 'Type A Cargo', whilst Cobalt, Lithium, Rare Earths, Supplies, and Titanium were relatively less valuable, hence classified as 'Type B Cargo'.

Workers' Compensation's EDA evidenced variation driven most predominately by occupation. As such, this hazard area was split into the sub-categories, White Collar, Exploration Operations, Extraction Operations, and Spacecraft Operations. Finally, unlike the other hazards, it was concluded not to split the business interruption dataset any further as there were no significant variables that assisted in pricing nor warrant their own pricing categories (refer to *Appendix A* for a detailed explanation). Intriguingly, even variables such as safety compliance and maintenance frequency held minimal importance. For example, a level 5 safety compliant operation is equally compelled to cease business operation during a fierce solar storm (or any other adverse solar event) as a level 1 compliant operation.

## 2.1 Equipment Failure

### 2.1.1 Benefit Structure

Equipment Failure insurance provides coverage for losses arising from mechanical breakdown, electrical failures, explosions and operational malfunctions during the mining process. The benefit structure provides compensation for the repair or replacement of the insured equipment upon failure, with payout amounts based on the condition of the equipment at the time of claim. To ensure consistency of estimates and revenue sustainability, coverages are capped at 790K per claim, limiting exposure to overly large losses. Premiums for 'High Risk' equipment were priced higher than those for 'Low Risk' equipment to ensure losses were accurately accounted for. The characteristics of the covered solar systems were also considered, with their risk factors being evaluated based on potential damage to the insured equipment. This ultimately led to higher premiums for the Bayesia System, lower premiums for the Helionis Cluster, and standard base premiums for Oryn Delta, which had no major risk attributes that would warrant a higher premium price.

### 2.1.2 Coverage Triggers and Exclusions

The coverage triggers of the Equipment Failure insurance are predominantly from unexpected events that occur non-intentionally, which result in unforeseen physical damage that prevents the machinery from operating normally. For a claim to be processed, evidence of damage or breakdown must be provided to ensure claim validity. Additionally, all insured equipment must meet the initially required safety standards. To ensure healthy profits and maintainable products, claims arising from the following reasons are not covered by the equipment failure policies: gradual wear and tear, intentional misuse, a lack of maintenance, not adhering to safety standards and any pre-existing faults known at the time of policy inception. Additionally, any equipment older than 30 years will not be eligible for insurance.

## 2.2. Cargo Loss

### 2.2.1 Benefit Structure

Cargo Loss insurance provides coverage for loss and damage to the cargo resulting from accidental damage, theft and environmental impacts during transportation. The coverage is provided on a per shipment basis, with a maximum claim amount of 38 million (Type B)

and 678 million (Type A). The indemnity is based on the declared value at policy inception, with the payout determined according to the extent of the loss during transportation. The policy covers the full value of the insured loss, up to the declared value. Cargo risks were found to vary across the solar system and will be assessed based on route and operating conditions. Premium adjustments were determined by considering specific route risks, transit duration, and hazards for operations in select solar systems, namely, Helionis Cluster and Oryn Delta.

#### 2.2.2 Coverage Triggers and Exclusions

The coverage is triggered from losses and damages occurring due to events such as collisions, navigation hazards, and environmental hazards such as asteroids and solar activity. Furthermore, coverage extends to third-party interference and operational failures. All events must occur within the insured period and be consistent with standard operating conditions.

Only Type B Cargo is covered, with *Type A Cargo excluded* due to its highly concentrated risk and potential loss. Additionally, non-accidental losses, delays without physical damage, and incidents occurring outside the insured period are not covered. All the routes used for transportation must meet established safety standards and achieve minimum clearance requirements to be eligible for coverage.

### 2.3. Workers' Compensation

#### 2.3.1 Benefit Structure

The Workers' Compensation hazard component of the insurance provides coverage and financial support for CQMC employees who sustain injuries, illness and work-induced conditions. The product provides payments for lost income when an employee is unable to work due to injury, and also covers medical, hospital, and rehabilitation costs. Coverage is based on a per-event basis with a cap of 170K per claim. Non-white Collar workers will be compensated for the physical injuries and illnesses arising from hazardous environments, while White Collar workers will be targeted towards psychological claims, including stress-related claims and other workplace factors. Additional considerations apply to employees operating within extreme solar system environments, most explicitly the Bayesia system. For example, the higher levels of radiation present and subsequent increased exposure will require workers to meet a relatively higher exposure threshold to be eligible in making radiation related claims. Conversely, those who work in lower pre-defined exposure systems will be eligible for potential discounts.

#### 2.3.2 Coverage Triggers and Exclusions

Coverage will only be triggered from claims for injury or illness sustained during the course of employment with the firm, with potential causes including workplace accidents, environmental hazards, operational incidents and work-related stress. The claims will be verified through evidence, such as medical examinations, to ensure valid claims are paid out. There will be exclusions applied to limit the amount and frequency of the claims. For example, intentional injuries, self-inflicted injuries, pre-existing medical conditions and death will not be covered as part of this insurance policy.

### 2.4. Business Interruption

#### 2.4.1 Benefit Structure

Business Interruption insurance claims can be made to compensate for revenue loss and cover fixed operational expenses, including salaries, rent and utilities. The maximum payout amount per event is 1,426K to limit exposure to extreme claims. Premiums for the three solar systems were priced the same, as they all exhibited similar levels of interruption risk.

### 2.4.2 Coverage Triggers and Exclusions

The coverage triggers for business interruption policies involve large operational delays or forced shutdowns caused by unforeseen and unintentional external factors such as natural disasters, supply failure, burglary, satellite failure, or communication failure. Conditions that must be met for a valid claim include upholding proper maintenance requirements and providing proof of loss. Exclusions for business interruption claims include failure to maintain scheduled maintenance of communication systems, internal conflicts such as worker strikes, and structural issues such as bankruptcy, liquidity or financial problems. In the case of extreme events, claims for business interruption can only be collected for revenue loss. Any specific damages (e.g., to equipment) must be claimed through the other respective Galaxy General Insurance products.

## 3. Summary of Pricing and Capital Modelling

### 3.1 Modelling

Prior to pricing premiums for each hazard coverage, the appropriate distributions for severity and frequency were first identified to estimate aggregate losses. The severity distribution, which captures claim size, was selected from either the Gamma or Inverse Gaussian families. Whilst the frequency distribution, which models the number of claims, was chosen from Poisson, Negative Binomial, Zero-inflated Poisson and Zero-inflated Negative Binomial. The most suitable model was selected by including all candidate variables and assessing the model's performance based on the statistical metrics AIC and BIC.

Following the selection of the best-fitting distribution, a forward subset selection was conducted to determine the most relevant explanatory variables. These variables were then used to fit Generalised Linear Models (GLMs), with link functions corresponding to the chosen distributions shown in Table 1 (*refer to Appendix B for full model specifications*).

| Hazard Coverage | Frequency | Severity |
|---|---|---|
| **Equipment Failure** | Negative Binomial | Gamma |
| **Cargo Loss (Type A)** | Poisson | Gamma |
| **Cargo Loss (Type B)** | Negative Binomial | Gamma |
| **Worker's Compensation** | Negative Binomial | Gamma |
| **Business Interruption** | Negative Binomial | Inverse Gaussian |

**Table 1: Model Distributions**

Finally, a Monte Carlo simulation with 10,000 iterations was conducted to estimate the aggregate loss distribution. The number of simulations was chosen with considerations to both computational constraints and accuracy. Under each iteration, the number of claims was first simulated using the frequency GLM, followed by a simulation of the individual claim size using the severity GLM. The total loss was then calculated by aggregating all simulated claims across the portfolio *(Appendix)*. Through this process, a distribution for aggregate loss was formed, which was then used to estimate expected loss, variance and the tail behaviours.

| | Equipment Failure | Cargo Loss (Type A) | Cargo Loss (Type B) | Workers' Compensation | Business Interruptions |
|---|---|---|---|---|---|
| **Expected Value** | 47,824,099 | 214,759,092,951 | 8,131,007,785 | 3,953,181 | 47,068,079,272 |
| **Variance** | 1.08E+14 | 1.70E+19 | 2.72E+16 | 2.96E+11 | 9.91E+17 |
| **VaR_90%** | 59,968,164 | 220,024,910,071 | 8,341,075,937 | 4,642,532 | 48,310,348,832 |
| **VaR_95%** | 64,890,134 | 221,605,942,519 | 8,396,109,449 | 4,848,489 | 48,685,684,313 |
| **VaR_99%** | 75,651,712 | 224,514,388,783 | 8,531,384,245 | 5,246,498 | 49,465,622,831 |

**Table 2: Model Results**

### 3.2 Pricing

Using the aggregate loss values simulated, the short-term and long-term cash flows were projected. Using industry standards, a cost flat rate of 10% was applied to all hazard

areas, capturing administration, processing, and tax costs. Further, with considerations to specific characteristics, the hazard areas were segmented and assigned respective loading premiums to reflect a more accurate pricing guide. Table 3 represents the one year (2175) financial outcomes for each hazard area. Total Cost, calculated via the aforementioned simulated aggregate loss, is used to derive the Pure and Gross Premiums for each hazard area and their respective sub-sections. The Premium Income represents the sum of the total gross premium amounts, while Investment Income assumes the entirety of the Premium Income is invested into a well-diversified portfolio at the 2174 interest rate. As such, the inclusion of Total Cost, Premium Income, and Investment Income determines the Profit and Profit Margin.

| | Equipment Failure | Cargo Loss (Type B) | Workers' Compensation | Business Interruption |
|---|---|---|---|---|
| **Total Cost** | $47,824,099 | $8,131,007,785 | $3,945,460 | $47,068,079,272 |
| **Premium Income** | $56,334,895 | $9,454,660,215 | $4,852,302 | $55,374,210,908 |
| **Investment Income** | $2,873,438 | $482,247,882 | $247,498 | $2,824,084,756 |
| **Profit** | $11,384,234 | $1,805,900,312 | $1,154,340 | $11,130,216,393 |
| **Profit Margin** | 20.21% | 19.10% | 24% | 20.10% |

**Table 3: Short Term Predicted Financials**

Across the four hazards, profit margins remain relatively consistent between 19% - 24%, with Workers' Compensation performing best despite being the smallest line by premium volume. Business Interruption and Cargo Loss dominate the portfolio in scale, collectively accounting for over 99% in total premium. For Cargo Loss and Business Interruptions, it should be noted that CMCQ will realistically not be able to afford the total premium amount given their 2174 Net Revenue of $61bn. It is assumed that they will purchase individual policy units, whereby the number of policy units purchased is determined at the discretion of CQMC. However, as the insurer, Galaxy General must consider the scenario in which all potential units are bought. This assumption is carried on to the pricing of reserves and subsequent profit margins.

Long term financial expectations were also projected for the next 10 years with several key assumptions. Given the 10-year expansion rates for the 3 solar systems are 25%, 25% and 15%, the average one-year expansion rate was assumed to be 2.2%. Inflation for the next 10 years was projected using a 3-year moving average to help capture recent economic trends whilst smoothing out short-term volatility. Similarly, interest rates for 10 years were projected using a 5-year moving average to help capture longer term market structure. To account for the growth in operations, the number of policies were projected to increase at a rate of 1%, 0.5%, 1%, and 0.5% per year for Equipment Loss, Cargo Type B, Workers Compensation, and Business Interruptions respectively. Finally, the reserves were calculated for each year. Assuming that no reserves are held previously, 50% of the total profits at the end of 2175 are invested in 2176 whilst the other 50% are held in reserves. Investment funds are then accumulated annually, supplemented by an additional 95% of the total premiums received at the beginning of each year, with the remaining 5% allocated to reserves.

Aggregate Loss projections for all four hazard areas seem to follow a similar, consistent upward trend with an increase of approximately 70% over the 10 years. Profit margins also tend to improve sharply in the early years and follow a steady upward trajectory thereafter.

### 3.2.1 Equipment Failure

Evidently, while high-risk units carry greater per unit loss potential, the bulk of aggregate premium income is generated by the low-risk cohort.

| Equipment Classification | Number of Equipment | Loading Premium | Gross Premium | Total Premium |
|---|---|---|---|---|
| High Risk | 1265 | 8% | $12,330 | $15,597,763 |
| Low Risk | 3465 | 4% | $11,757 | $40,737,131 |

**Table 4: Premium Results for Equipment**

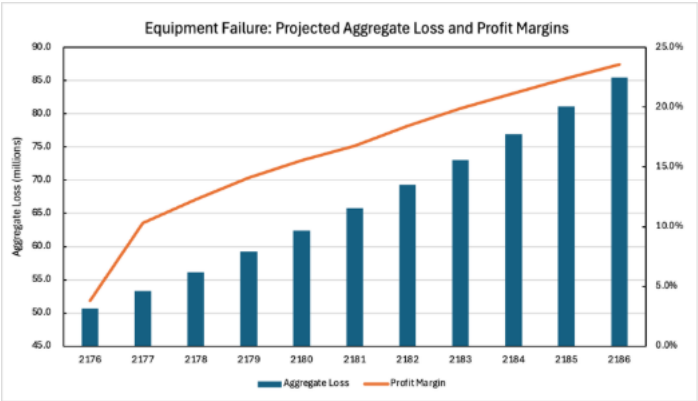



*Figure 1: 10-year Projected Financials for Equipment*

### 3.2.2 Cargo Loss (Type B)

Type B Cargo Loss represents an undifferentiated risk classification with a 4% loading premium.

| Classification | Loading Premium | Gross Premium | Total Premium |
|---|---|---|---|
| Type B | 4% | $1,128 | $9,454,660,215 |

**Table 5: Premium Results for Cargo**

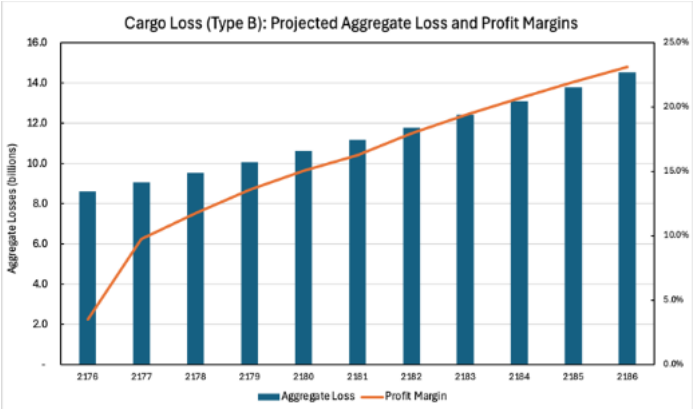



*Figure 2: 10 Year Projected Financials for Cargo Loss (Type B)*

### 3.2.3 Workers' Compensation

The Workers' Compensation pricing model is segmented into four different classifications of workers, with loading premiums calibrated in reference to the EDA and Safe Work Australia (2023) industry standards.

| Workers Classification | Number of Workers | Loading Premium | Gross Premium | Total Premium |
|---|---|---|---|---|
| White Collar | 6003 | 1% | $126.66 | $760,340 |
| Exploration Operations | 1205 | 6% | $134.20 | $161,710 |
| Extraction Operations | 19790 | 10% | $140.91 | $2,788,595 |
| Spacecraft Operations | 8811 | 3% | $129.57 | $1,141,657 |

**Table 6: Premium Results for Workers' Compensation**

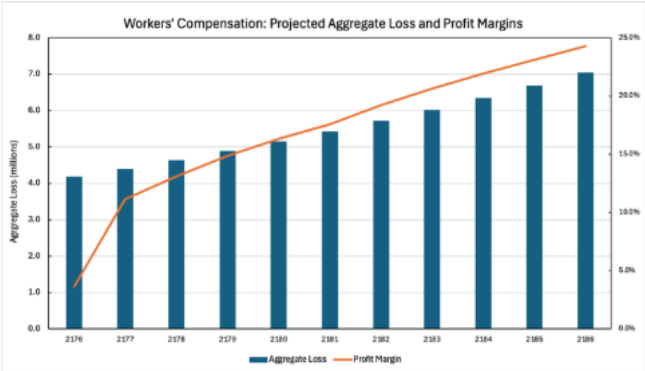



*Figure 3: 10 Year Projected Financials for Workers' Compensation*

### 3.2.4 Business Interruption

Business Interruption is treated as a single classification. The resulting total premium of $55.4 billion makes this hazard the largest line of income among all the insurance products.

| Classification | Loading Premium | Gross Premium | Total Premium |
|---|---|---|---|
| **Business Interruption** | 5% | $562,038 | $55,374,210,908 |

**Table 7: Premium Results for Business Interruption**

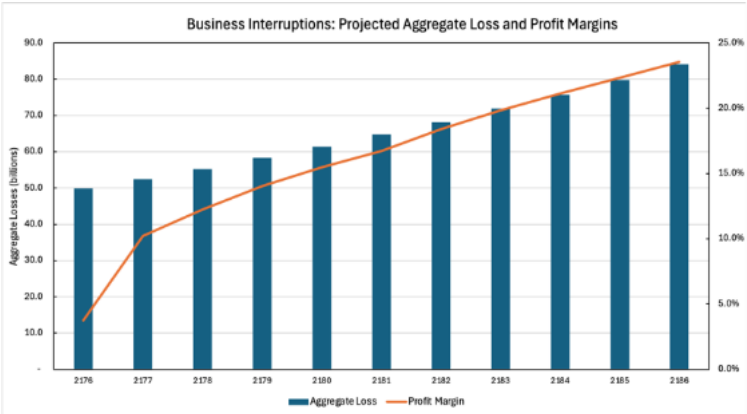



*Figure 4: 10 Year Projected Financials for Business Interruptions*

### 3.3 Stress Testing for Extreme Scenarios

Furthermore, the team has conducted sensitivity testing to evaluate the impact of extreme scenarios, including both a lower claims environment and severe, correlated events associated with solar system shocks. These scenarios were tested with results summarised in Table 8. An exemption was made for Cargo, where solar system correlation effects were not applied due to the lack of system-specific data. As such, two cases were tested: the first, where only frequency was inflated by 1.5; and the second, where both frequency and severity were increased by 1.5. To reflect the best-case scenario for all four hazard areas, claim frequency was lowered by a factor of 0.8. This resulted in lower aggregate losses and consequently improved projected profit margins. In contrast, to capture the worst-case scenarios, three progressively adverse scenarios were tested for all hazards (excluding Cargo), to reflect the impact of incrementally increased solar correlation, with the following assumptions:

- 1 Solar Correlation - 1 solar system being adversely affected;
- 2 Solar Correlations - 2 solar systems are adversely affected; and
- 3 Solar Correlations - A severe and highly correlated stress environment where all 3 solar systems are impacted (rare but possible).

Under all these scenarios, both the claim frequency and severity were inflated by 1.5, reflective of a severe shock, in turn capturing tail risk more robustly. The chosen scenarios and their respective inflation factors, allow for the analysis of correlation-driven events, without venturing into an unrealistic domain.

| Equipment | | | | | |
|---|---|---|---|---|---|
| Scenario | Best Case | Moderate Case | 1 Solar Correlation | 2 Solar Correlation | 3 Solar Correlation |
| Mean Aggregate Loss | 38,265,347 | 47,824,099 | 71,473,251 | 90,389,501 | 107,457,419 |
| 95% VaR | 53,490,238 | 64,890,134 | 101,991,703 | 121,508,828 | 139,142,118 |
| 99% VaR | 62,898,943 | 75,651,712 | 121,980,865 | 139,185,043 | 157,238,831 |
| Profit Margin Impact | 13.42% | 3.78% | -12.35% | -27.02% | -40.27% |
| Years to Recover | - | - | 2 years | 3 years | 3 years |

| Worker's Compensation | | | | | |
|---|---|---|---|---|---|
| Scenario | Best Case | Moderate Case | 1 Solar Correlation | 2 Solar Correlation | 3 Solar Correlation |
| Mean Aggregate Loss | 3,156,698 | 3,953,181 | 5,008,028 | 7,116,321 | 8,868,728 |
| 95% VaR | 3,953,059 | 4,848,489 | 6,167,212 | 8,542,592 | 10,491,509 |
| 99% VaR | 4,347,276 | 5,246,498 | 6,727,565 | 9,225,543 | 11,258,155 |
| Profit Margin Impact | 13.03% | 3.67% | -3.88% | -23.13% | -39.13% |
| Years to Recover | - | - | 1 years | 2 years | 3 years |

| Bussiness Interruption | | | | | |
|---|---|---|---|---|---|
| Scenario | Best Case | Moderate Case | 1 Solar Correlation | 2 Solar Correlation | 3 Solar Correlation |
| Mean Aggregate Loss | 37,670,947,059 | 47,068,079,272 | 59,455,598,728 | 81,749,099,770 | 105,934,088,378 |
| 95% VaR | 39,190,321,772 | 48,685,684,313 | 61,842,355,852 | 84,878,626,568 | 109,582,221,420 |
| 99% VaR | 39,775,775,997 | 49,465,622,831 | 62,959,870,363 | 86,497,389,135 | 111,344,640,303 |
| Profit Margin Impact | 13.40% | 3.76% | -3.80% | -21.40% | -40.50% |
| Years to Recover | - | - | 1 years | 2 years | 3 years |

| Cargo Type B | | | | |
|---|---|---|---|---|
| Scenario | Best Case | Moderate Case | Freq x 1.5 | Freq and Sev x 1.5 |
| Mean Aggregate Loss | 6,507,638,129 | 8,131,007,785 | 12,191,243,709 | 18,286,865,564 |
| 95% VaR | 6,745,955,787 | 8,396,109,449 | 12,554,348,925 | 18,831,523,387 |
| 99% VaR | 6,864,176,542 | 8,531,384,245 | 12,736,828,832 | 19,105,243,248 |
| Profit Margin Impact | 13.29% | 3.51% | -13.05% | -41.31% |
| Years to Recover | - | - | 2 years | 4 years |

**Table 8: Scenario Test Results**

It is evident that a potential 1 in 10-year loss event results in an initial, sharp drop in profit margins across all hazard areas as depicted in Table 8. However, with the assumption that premiums and business growth remain consistent, and the effects of catastrophic events do not carry onto subsequent years, profits will be able to recover quickly due to the strong backup of reserves, even in the most adverse scenarios. For example, if the catastrophe were to occur in the year 2176, when reserves are at their lowest and most vulnerable to extreme shocks, testing suggests that the reserves and profits will be able to recover within four years (*Appendix C1*). Despite the initial presumption that the worse-case scenario occurs in the year 2176, additional stress testing validates that the business will always be able to recover within ten years, regardless of when in the projection period the scenario occurs.

## 4. Risk Assessment

### 4.1 Risk Matrix

The risk matrix highlights that most of the primary risks have a low to moderate likelihood of occurrence. However, some risks can have a high and lasting impact on product performance. The top threats of Economic & Financial risk and Catastrophic risk are noticeably external perils, leaving Galaxy General more exposed and vulnerable. Interest Rate risk is also considerably high given the reliance on investment income to generate profits and capital. To reduce the impact, mitigation strategies have been outlined in Section 4.2 to ensure that the products remain profitable, robust and useful.

**Figure 5: Risk Matrix**

### 4.2 Tabulation of Risk Types

| Risk Type | Description | Mitigation Strategy |
|---|---|---|
| Catastrophic Risk | A correlated catastrophe event could trigger simultaneous large claims across all four coverages and all three systems. | Implementation of policy limits to cap the maximum exposure from a single correlated event. This is particularly relevant for the Cargo dataset, whereby catastrophe risk is high, and losses tend to be extreme. Also, a proportion of premiums will be set aside as reserves to account for potential catastrophic events. |
| Data and Model Risk | The absence of credible system specific claims data means the models rely heavily on assumptions. | Ensure assumptions are realistic, coherent and grounded by some data and/or factual evidence. Ensure models are validated by common metrics including, AIC and BIC. Models will be updated with new data regularly to ensure they remain reliable and accurate. |
| Operational Risk | Communication delays between solar systems slowing claims detection and response. | Implement an intergalactic standard procedure for acceptable time frames for claim detection and response. Impose a buffer period in case of unforeseen circumstances (e.g., a 72-hour claim extension). Adhere to contingency protocols. |
| Underwriting Risk | Risk that the actual claims deviate adversely from priced expectations. | Consistently monitor and review figures annually and use emerging claims to ensure the product remains fit for purpose. |
| Regulatory Risk | Interstellar insurance is regulated under the Galactic Insurance Authority with compliance obligations and capital requirements. | Maintain ongoing engagement with the Galactic Insurance Authority (GIA) to monitor evolving compliance obligations. Capital reserves should be held above the minimum GIA benchmark to buffer against future regulatory changes. |
| Economic & Financial Risk | Economic recessions may increase lapse rates, cause higher inflation and unexpectedly reduce demand for insurance products. | Make economic projections to determine economic cycles (e.g. boom or bust) and plan accordingly. Hold additional reserves during inflationary periods and similarly, ensure strong cash inflows during recessional periods in case of high policy lapse rates. |
| Competition Risk | Uncompetitive premiums may reduce market share and jeopardise profitability. | Ensure premium loading factors are consistent with competitors. Ensure products are priced competitively. Regularly monitor premium prices within the industry and respond accordingly. |
| Interest Rate Risk | Investment income may decline if interest rates fall below expectation or vary erratically. | Construct a portfolio that combines interest rate sensitive assets with less sensitive assets to reduce volatility. |
| Ethical Risk | Pricing may unintentionally bias different individuals based on certain (policyholder) characteristics. | Risk pooling is used to spread the financial burden of losses across all individuals. For example, pool the risks of *drill operators* and *C-suite executives*. This ensures equitability. |

## 5. An Evaluation of Ethics and ESG Considerations

The Galaxy General Insurance Company has a strong commitment to upholding ESG values, which is underpinned by strong ethical guidelines. A prominent ethical consideration entails ensuring the affordability of insurance products for all socioeconomic classes. For example, drill operators are exposed to significant risk in their daily duties relative to technical employees. The foundational principle of risk-pooling is used to help distribute aggregate risk across the policyholder pool. Otherwise, premiums would be more exorbitant for the lower-skilled workers, who tend to come from lower socioeconomic classes, contributing to greater income inequality. Furthermore, CQMC's operations are, by nature, detrimental to the solar system environment. However, such operations have been necessitated due to the depletion of resources on Earth and providing insurance for these intense mining operations and expansion plans are ethically justified by the primitive need for survival. As an insurer, climate risk will be regularly reviewed to ensure the suitability of individual product design. Furthermore, strong corporate governance procedures have been implemented to ensure the financial feasibility of the insurer's operations (i.e. ensuring solvency). This includes setting aside strong reserves, stress testing and profit testing various possible catastrophe events.

## 6. Assumptions

General report assumptions are based on knowledge of the current earthly environment due to a lack of future understanding. Further, it was assumed that the historical claims data for the three existing solar systems can be used as a proxy and be accurately mapped onto the emerging solar systems, given that this was the only data available. It is assumed that CQMC will have consistent yearly mining yield for the given time horizon. As more data is acquired, yearly revisions of product design, modelling, pricing and key assumptions will be made, in line with the 'monitoring results' component of the ACC. Specific assumptions and their rationale for product design, modelling, and pricing can be found in their respective subsections (above).

## 7. Data and Data Limitations

The claims datasets contained several limitations that required remediation prior to modelling. Observations with values falling outside the given, plausible range of each variable were removed as it was likely a data entry or recording error that cannot be inferred. A special case was applied for the Equipment dataset, where only negative "equipment ages" were removed since otherwise a significant proportion (42,000) claims would have needed to be removed. The values of claims for Workers Compensation were inferred to be in "K's" (thousands). Certain categorical fields contained irregular formatting, such as non-standard characters and transcription errors, which were standardised to ensure consistent grouping. Further, policies that presented a 0-claim count with a positive claim amount were removed. Some records contained missing values, however there was sufficient contextual information across the severity and frequency sheets for this data to be reconciled. The data cleaning decisions had a minimal overall impact on the data with 1.24%, 5.8%, 1.35%, 1.4% of the four ordered hazard datasets being removed respectively. However, their impact on modelled loss estimates is recognised and is still considered in the interpretation of results.

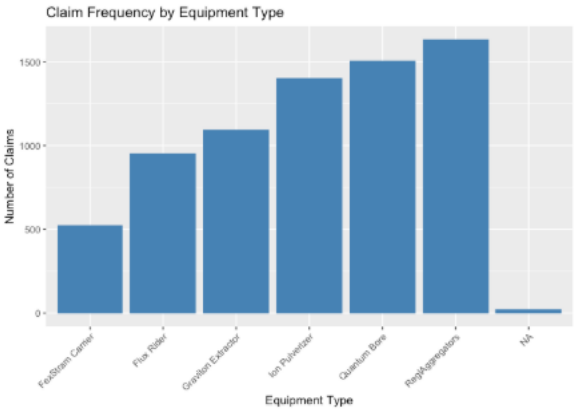
## Conclusion

Overall, it is recommended that Galaxy General sells the four insurance products excluding Cargo Type A to all three solar systems with tailored product designs. This insurance undertaking is financially sustainable in the long term despite the potential exogenous risks posed by the expansion into new solar systems.

# Appendix

## Appendix A: Exploratory Data Analysis (EDA): Visualisations & Key Insights

### Equipment (Key Variables):



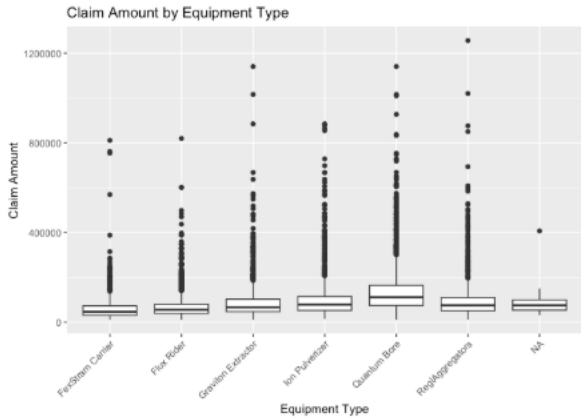
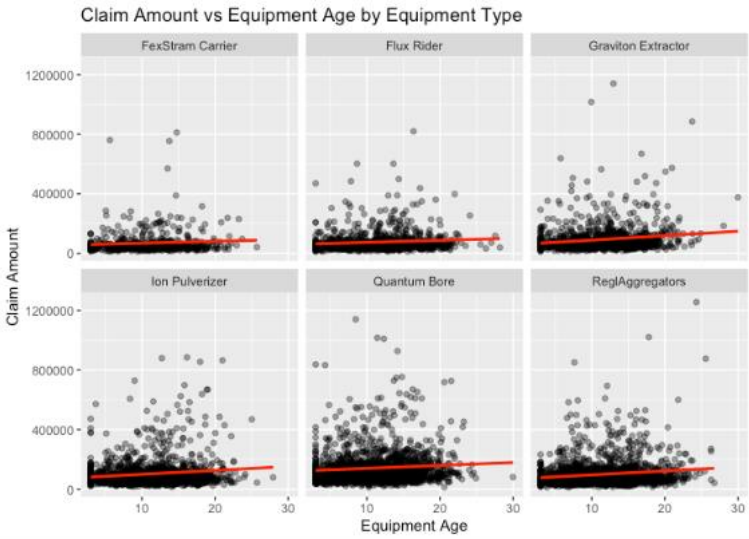



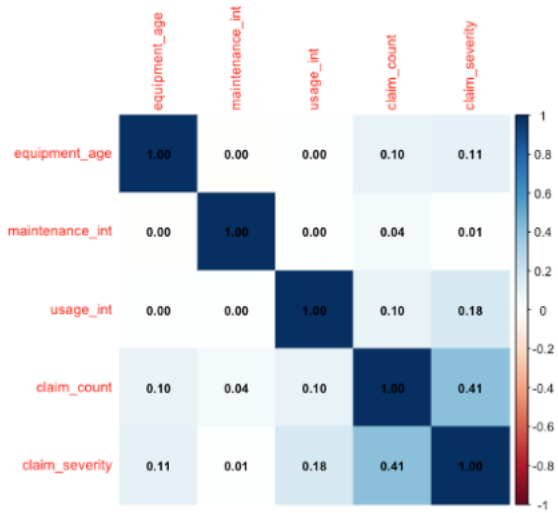

***Figure 1A: Claim Amount by Equipment*** ***Figure 1B: Claim Frequency – Equipment***

- In Figure 1A, Quantum Bore tends to have the highest average claims across all equipment types
- But Regl Aggregators tend to have the highest frequency claim amount in Figure 1B
- The level of claims and frequency was consistent across the solar systems



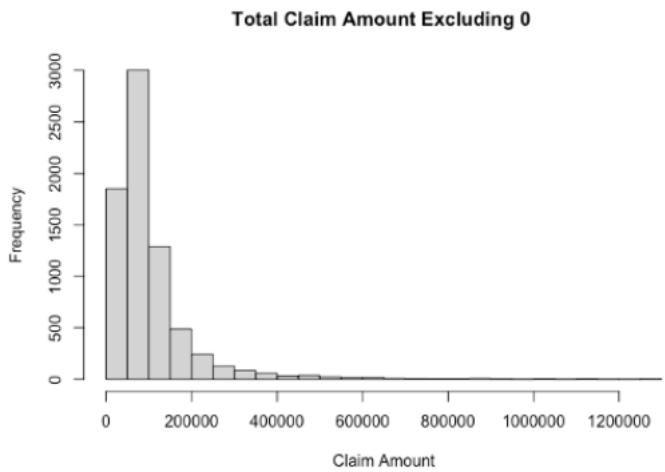

***Appendix Figure 1C: Claim Amount against Equipment Age by Equipment Type***

- Figure 1C, most equipment failures produce small or moderate losses
- A few events create very large claims (outliers)
- Catastrophic failures are not limited to old equipment
- The red lines are not steep - so there is weak relationship between age and equipment type
- In general, as well age does not greatly affect the claim amount and no prominent degree of relationship by solar system




***Appendix Figure 1D: Correlation Matrix***

- Figure 1D emphasises that there is no strong relationship present amongst the variables, with the 0.18 amongst usage intensity and claim severity
- The positive correlation of 0.11 confirms the previous relationship determined with claim severity and equipment age
- The above matrix further highlights, there is lack of correlation amongst variables themselves, with values close to 0, removing concerns of multicollinearity

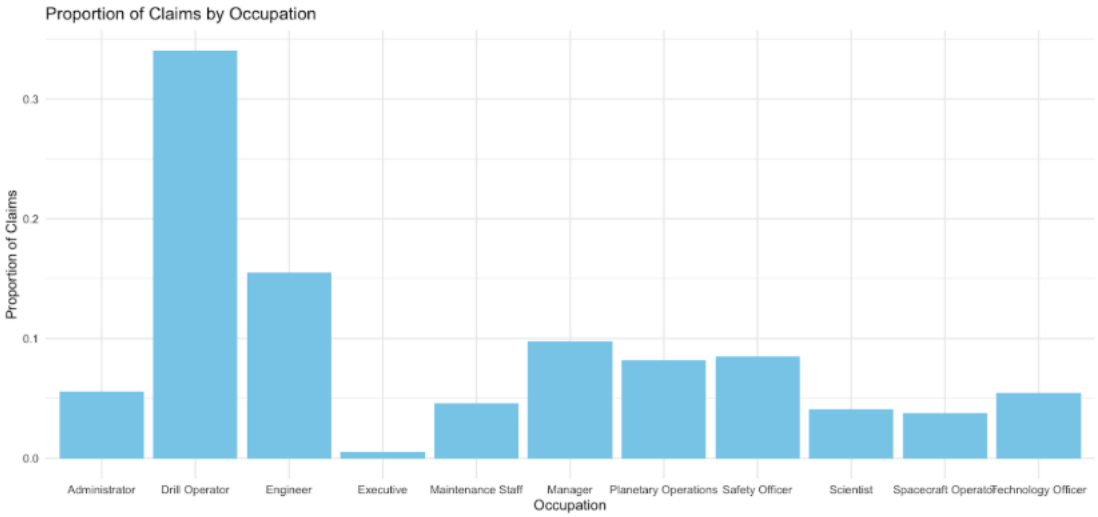



***Appendix Figure 1E: Histogram of Claim Amount***

- Appendix Figure 1E, shows the claims are highly right skewed, with a long tail with outliers in the dataset
- Most values are concentrated in towards the lower amounts between 0 -150,000 (lowest claim bins)

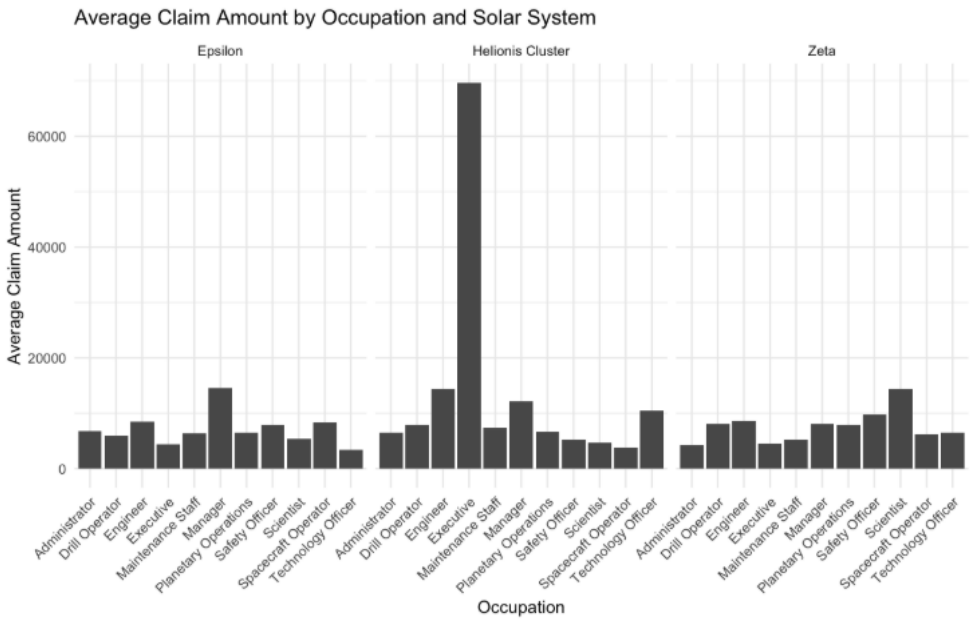
## Workers' Compensation (Key Variables)

Overall, note that all variables were analysed but seen to have minimal or no strong driving relationships in affecting outcomes.



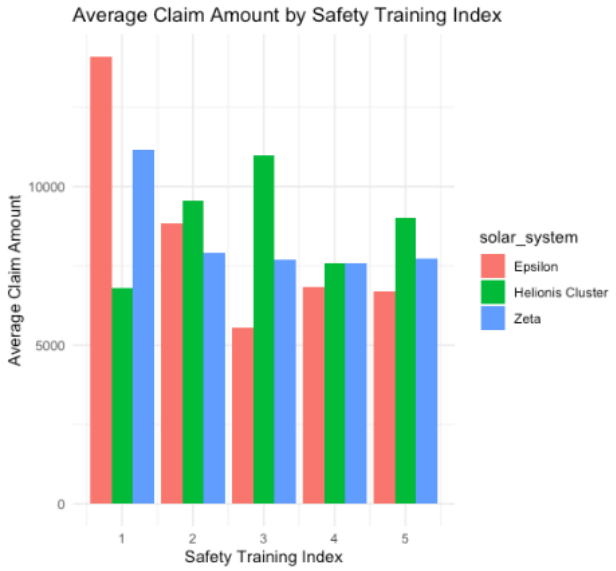

***Appendix Figure 2A: Proportion of Claims by Occupation***

- Drill operators have significantly higher proportion of claims, followed by engineers - this can be due to the high-risk nature work
- This trend holds across all solar systems and overall, in general




***Appendix Figure 2B: Average Claim Amount by Occupation and Solar System***

- Interestingly, when looking at average claim amounts, drill operators and engineers are similar for HC and EP, reflecting the risky nature work
- Meanwhile, for Zeta, drill operators have a significantly larger average claim amount.
- There was an outlier for Executive under Helionis Cluster, which inflated the average claim amount

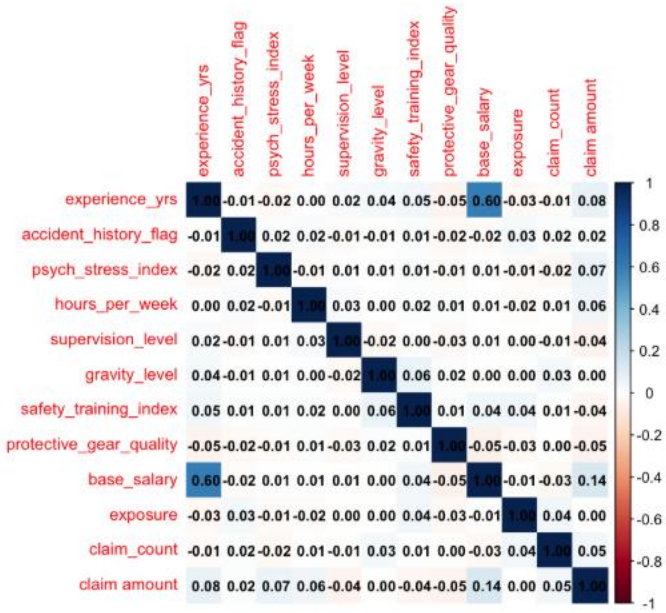



***Appendix Figure 2C: Average Claim Amount by Training Index***

- Higher safety training levels appear to be associated with lower claim severity, suggesting improved worker training may reduce the severity of workplace incidents.
- For Epsilon, high claim amounts were observed for workers with a safety training index of 1, with a decreasing trend as safety training increased. This highlights the importance of effective safety training programs to mitigate risky workplace behaviour.
- The graph reveals a more volatile pattern for Helionis Cluster, where workers with a safety training index of 3 resulted in the largest average claim amounts.
- The average claim amount remained relatively consistent across the different safety training indices for Zeta, with the exception of workers with a safety training index of 1, who had significantly higher claim amounts.

***Appendix Figure 2D: Claim Frequency***

- The total claim count increases as the safety training index increases
- This is interesting, as intuitively the expected you would lower claim frequency under lower ranked safety training index, as there is higher risk associated
- But in comparison to Figure 2C, whilst workers classified with higher safety index, tend to claim on average lower amounts

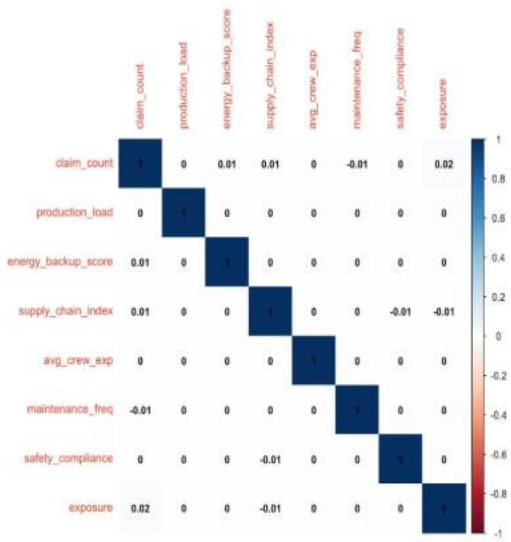



**Appendix Figure 2E: Correlation Matrix**

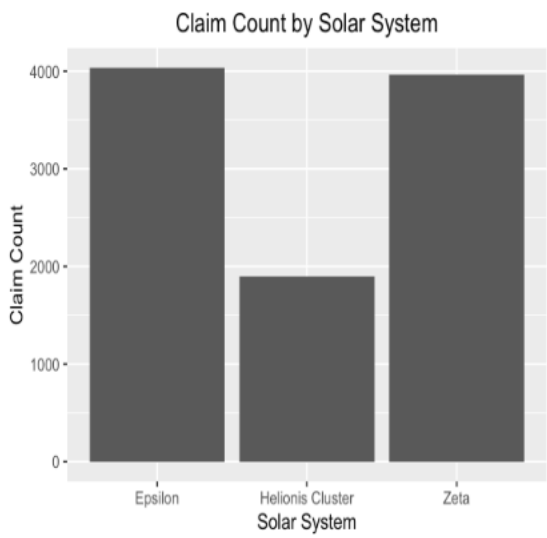
- Based on above matrix, there are no prominent strong positive relationships/ no significant correlation amongst most variables, limiting concerns of multicollinearity
- According to results, 0.14 relationship between base salary and claim amount
- Additionally, base salary was shown to have a moderate positive correlation with years of experience. This is quite intuitive given that greater experience usually translates to higher pay rates.

## Business Interruption:

### Frequency Analysis:




*Appendix Figure 3A: Correlation Matrix*




*Appendix Figure 3B: Claim Count by Solar System*

**Appendix Figure 3A Remarks:**

- Essentially every variable has a non-significant pairwise correlation.
- Exposure has a positive correlation with claim count. This is reasonable, as higher exposure to risk therefore results in an elevated chance of making a claim. However, this correlation value, while positive, is interestingly lower than anticipated.
- The correlation matrix suggests that more frequent maintenance is associated with a lesser likelihood of making a claim. This is logically sound but one would expect this negative relationship to be stronger than what is suggested.
- The variables *maintenance frequency, supply chain index* and *energy back-up score* were deemed to have some correlation to claim count and thus warrant further investigation in the modelling process of frequency.

**Appendix Figure 3B Remarks:**

- Epsilon has the highest claim count among all the three solar systems, just exceeding a total of 4000 claims.
- Zeta trails closely behind Epsilon in terms of total claim count, falling just short of 4000 claims.
- Helionis Cluster has approximately half the number of claims as Epsilon and Zeta.

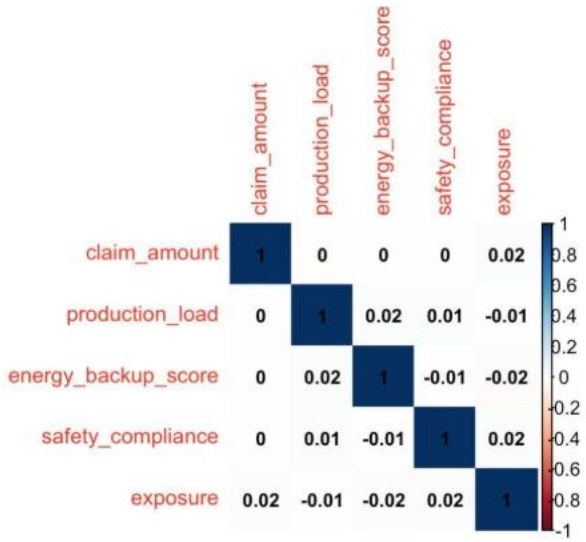
***Appendix Figure 3C: Claim Count by Safety Compliance***

**Appendix Figure 3C Remarks:**

- This validates the claim made under *Product Design* (Section 2) in relation to pricing. That is, the business interruptions dataset was not split into any sub-categories for pricing (unlike the other three datasets).
- Naturally, a good candidate for pricing would have been the variable *safety compliance*. That is, low safety compliance represents high risk and high safety compliance represents low risk. As such, premiums would have been allocated to their relative risk levels.
- The above graph, however, demonstrates that higher safety compliance does not appear to lessen the number of claims made. Intuitively, one would expect that lower

safety compliance would correspond to higher total claims and vice versa. Interestingly, such a trend is not evident, rather, all the safety compliance categories seem to elicit a relatively similar number of claims. This reiterates the fact that business interruptions seem to have a relatively uniform effect across the board, despite individual characteristics.

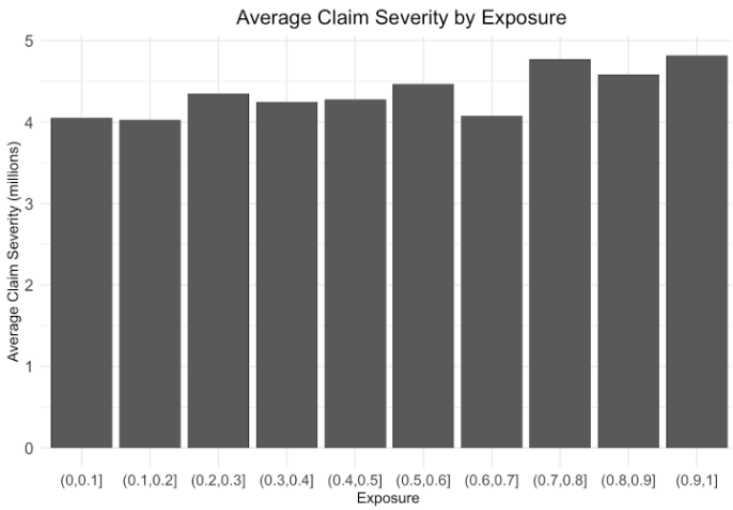
**<u>Severity Analysis</u>:**




***Appendix Figure 3D: Correlation Matrix (severity)***

**Appendix Figure 3D Remarks:**

- Once more, the pairwise correlation values appear to be non-significant for all the variables (similar to the previous correlation matrix for the frequency variables). *Exposure* has a positive correlation to claim amount.
- Interestingly, production load has zero correlation to claim amount. Intuitively, one would assume that a higher production load would correspond to a higher claim amount, given the elevated risk of product damage.
- *Exposure* should be investigated further in relation to statistical significance when modelling severity.



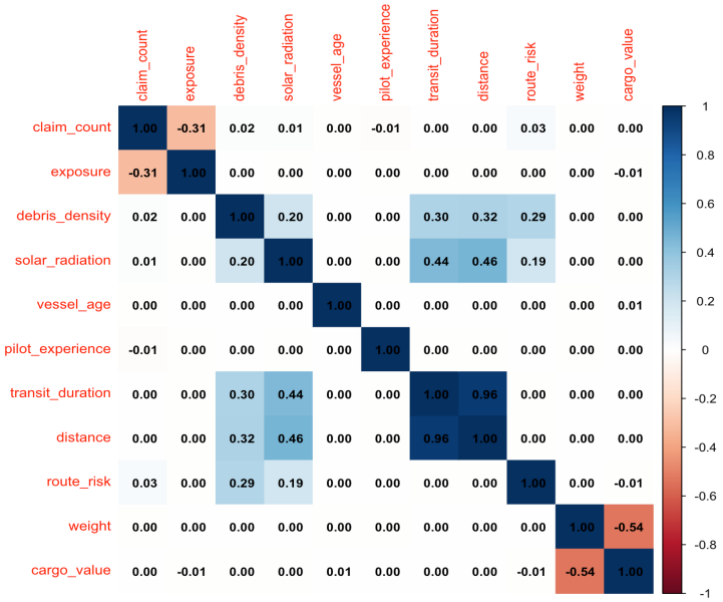

***Appendix Figure 3E: Average Claim Severity by Exposure***

**Appendix Figure 3E Remarks:**

- Higher exposure categories have a higher average claim severity. This is observed most apparently in the categories: (0.7, 0.8], (0.8, 0.9] and (0.9, 1]. These exposure categories have an average claim severity greater than $4.5 million.
- The low to middle exposure categories has an average claim severity between $4 - $4.5 million.
- This affirms the previous finding (from the correlation matrix) that greater exposure is associated with slightly higher claim severity.

## Cargo Loss:

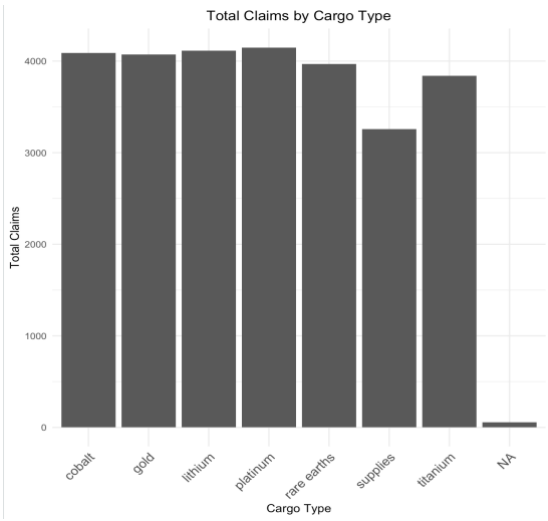
**Frequency**




| | claim_count | exposure | debris_density | solar_radiation | vessel_age | pilot_experience | transit_duration | distance | route_risk | weight | cargo_value |
|---|---|---|---|---|---|---|---|---|---|---|---|
| claim_count | 1.00 | -0.31 | 0.02 | 0.01 | 0.00 | -0.01 | 0.00 | 0.00 | 0.03 | 0.00 | 0.00 |
| exposure | -0.31 | 1.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | -0.01 |
| debris_density | 0.02 | 0.00 | 1.00 | 0.20 | 0.00 | 0.00 | 0.30 | 0.32 | 0.29 | 0.00 | 0.00 |
| solar_radiation | 0.01 | 0.00 | 0.20 | 1.00 | 0.00 | 0.00 | 0.44 | 0.46 | 0.19 | 0.00 | 0.00 |
| vessel_age | 0.00 | 0.00 | 0.00 | 0.00 | 1.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.01 |
| pilot_experience | -0.01 | 0.00 | 0.00 | 0.00 | 0.00 | 1.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| transit_duration | 0.00 | 0.00 | 0.30 | 0.44 | 0.00 | 0.00 | 1.00 | 0.96 | 0.00 | 0.00 | 0.00 |
| distance | 0.00 | 0.00 | 0.32 | 0.46 | 0.00 | 0.00 | 0.96 | 1.00 | 0.00 | 0.00 | 0.00 |
| route_risk | 0.03 | 0.00 | 0.29 | 0.19 | 0.00 | 0.00 | 0.00 | 0.00 | 1.00 | 0.00 | -0.01 |
| weight | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 1.00 | -0.54 |
| cargo_value | 0.00 | -0.01 | 0.00 | 0.00 | 0.01 | 0.00 | 0.00 | 0.00 | -0.01 | -0.54 | 1.00 |

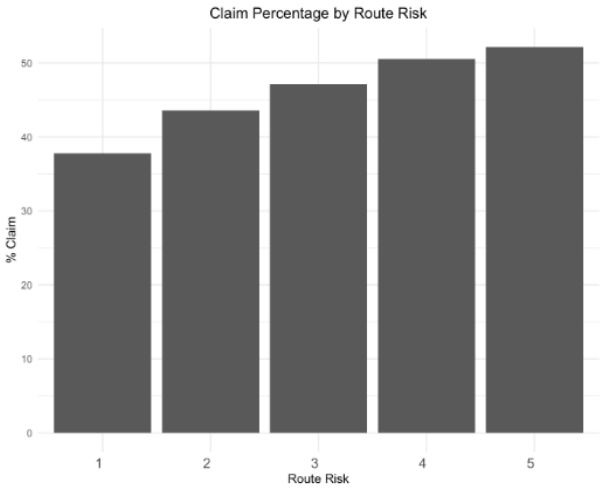
***Appendix Figure 4A: Correlation Matrix***

- Apparently, greater exposure will lead to fewer claim counts? (-0.31 correlation)
- Claim count has a trivial correlation with all other covariates.
- The longer the transit duration the more likely you are to encounter debris (or higher debris density). Corr = 0.30
- The longer the transit duration, the greater the exposure to solar radiation (corr = 0.44)
- The longer the distance, the higher exposure it has to solar radiation (0.46 cor)



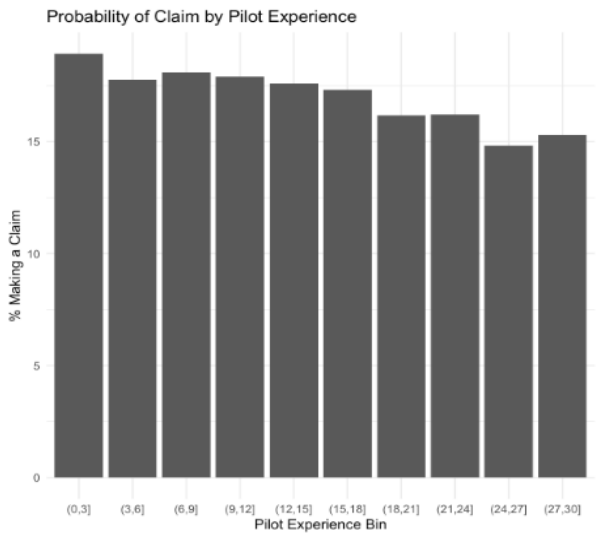

***Appendix Figure 4B: Total Claims by Cargo Type***

- All types of cargo have a relatively equal number of total claims. Interestingly, supplies is the least frequency category. Intuitively, one would expect more claims in supplies given perhaps the constant need for interplanetary travel.
- Otherwise relatively uniform across all types.




***Appendix Figure 4C: Claim Percentage by Route Risk***

- Higher route risk leads to a higher claim percentage (route risk of 4 and 5 has claim percentage >50%)
- On the other hand, route risk 1 has a claim percentage of ≈37%. Approx. 13% difference to route risks 4 and 5.

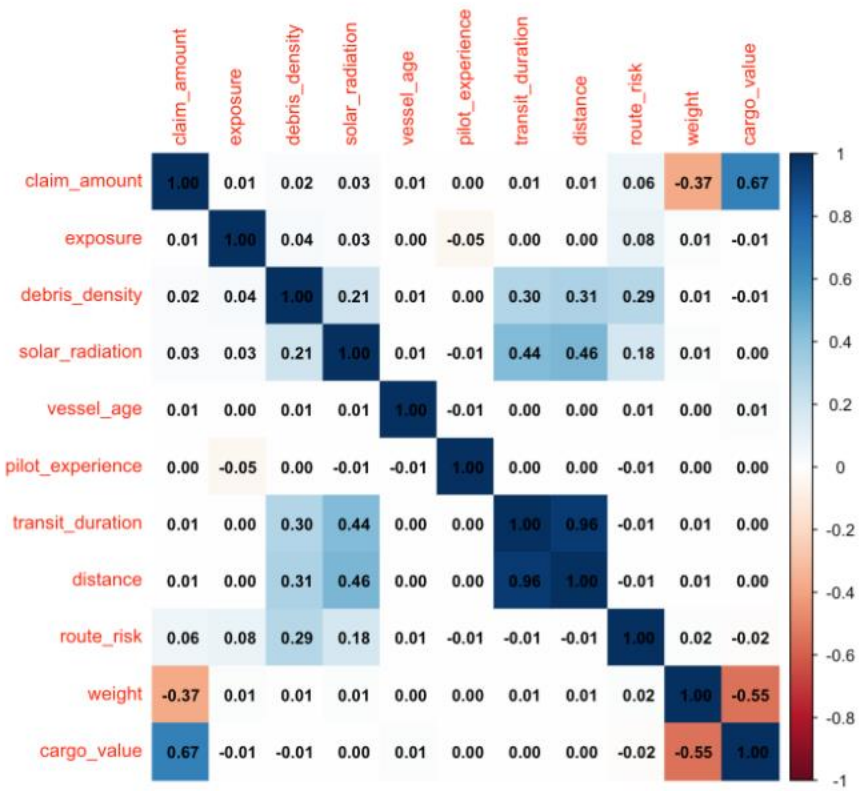



***Appendix Figure 4D: Probability of Claim by Pilot Experience***

- The more experience a pilot has, the less likely they are to make a claim.
- Plot experiences from [3,18] appear to be stable but thereafter a more noticeable reduction in % claims is observed.

**Appendix Figure 4E: Table of Summary of cargo type:**

| **Name** | **Value/Kg** | **Total claim count** |
|---|---|---|
| **Cobalt** | 52.97441 | 4087 |
| **Gold** | 133333.7 | 4073 |
| **Lithium** | 83.55867 | 4112 |
| **Platinum** | 53770.83 | 4148 |
| **Rare Earths** | 86.32001 | 3970 |
| **Supplies** | 10.16707 | 3257 |
| **Titanium** | 7.098377 | 3840 |




***Appendix Figure 4F: Correlation Matrix Severity***

- Investigate claim amount with cargo value (corr = 0.67)
- Investigate claim amount with route risk (corr = 0.06)
- There is lack of correlation between variables, limiting the concerns of multicollinearity in modelling

## Appendix B: Full Model Specifications (R Output)

### Business Interruptions

### Winning Frequency Model: Negative Binomial

```
Call:
glm.nb(formula = claim_count ~ offset(log(exposure)) + maintenance_freq +
    supply_chain_index + energy_backup_score + solar_system,
    data = freq_bus_clean, init.theta = 0.1108406848, link = log)

Coefficients:
                            Estimate Std. Error z value Pr(>|z|)
(Intercept)                -1.601490   0.049120 -32.603  < 2e-16 ***
maintenance_freq           -0.014851   0.007019  -2.116  0.03435 *
supply_chain_index          0.103412   0.048675   2.125  0.03362 *
energy_backup_score         0.030837   0.009934   3.104  0.00191 **
solar_systemHelionis Cluster -0.059332  0.038687  -1.534  0.12511
solar_systemZeta           -0.020798   0.031339  -0.664  0.50692
---
(Dispersion parameter for Negative Binomial(0.1108) family taken to be 1)

    Null deviance: 23958  on 97532  degrees of freedom
Residual deviance: 23938  on 97527  degrees of freedom
AIC: 62560

Number of Fisher Scoring iterations: 1


              Theta:  0.11084
          Std. Err.:  0.00276

 2 x log-likelihood:  -62545.53800
```

### Winning Severity Model: Inverse Gaussian

```
Call:
glm(formula = claim_amount ~ solar_system + exposure, family = inverse.gaussian(link = "log"),
    data = sev_bus_clean)

Coefficients:
                            Estimate Std. Error t value Pr(>|t|)
(Intercept)                 15.22251    0.04852 313.740   <2e-16 ***
solar_systemHelionis Cluster  0.08055    0.04823   1.670   0.0949 .
solar_systemZeta             -0.06607    0.03707  -1.782   0.0748 .
exposure                      0.14696    0.07884   1.864   0.0623 .
---
(Dispersion parameter for inverse.gaussian family taken to be 6.444061e-07)

    Null deviance: 0.0072289  on 9897  degrees of freedom
Residual deviance: 0.0072206  on 9894  degrees of freedom
  (8 observations deleted due to missingness)
AIC: 321045
```

***Note:*** *The Inverse Gaussian model was deemed to be superior relative to the Gamma Model, as evaluated by their respective AICs*. Inverse Gaussian AIC (321,045) < Gamma AIC (322,385). Similarly, Inverse Gaussian BIC (321108.1) < Gamma BIC (322,442.5).

## Equipment

### Winning Frequency Model: Negative Binomial

```
Call:  glm.nb(formula = claim_count ~ equipment_type + equipment_age +
    solar_system + maintenance_int + usage_int + offset(log(exposure)),
    data = freq_equi_clean, init.theta = 13.11712704, link = log)

Coefficients:
                     (Intercept)          equipment_typeFlux Rider  equipment_typeGraviton Extractor
                      -2.3269704                        -0.1266211                         -0.3712287
    equipment_typeIon Pulverizer        equipment_typeQuantum Bore     equipment_typeReglAggregators
                      -0.4232425                        -0.5658626                         -0.6664546
                  equipment_age      solar_systemHelionis Cluster                  solar_systemZeta
                       0.0098273                         0.3771192                         -0.0843606
               maintenance_int                          usage_int
                       0.0002261                         0.0408595

Degrees of Freedom: 93025 Total (i.e. Null);  93015 Residual
Null Deviance:      36000
Residual Deviance: 34700        AIC: 49160
```

### Winning Severity Model: Gamma

```
Call:  glm(formula = claim_amount ~ equipment_age + usage_int + equipment_type +
    solar_system, family = Gamma(link = "log"), data = sev_equi_clean)

Coefficients:
                     (Intercept)                    equipment_age                        usage_int
                       10.439234                         0.007369                         0.022751
        equipment_typeFlux Rider  equipment_typeGraviton Extractor      equipment_typeIon Pulverizer
                        0.143598                         0.366913                         0.492803
      equipment_typeQuantum Bore     equipment_typeReglAggregators      solar_systemHelionis Cluster
                        0.879207                         0.467124                        -0.098880
                 solar_systemZeta
                        0.120593

Degrees of Freedom: 8026 Total (i.e. Null);  8017 Residual
  (16 observations deleted due to missingness)
Null Deviance:      2791
Residual Deviance: 2001         AIC: 191800
```

*Note: that for Equipment, under AIC and BIC, Inverse Gaussian had better values but according to the residuals Gamma provided a better fit for the tail*

## Worker's Compensation

### Winning Frequency Model: Negative Binomial

```
Call:
glm.nb(formula = claim_count ~ offset(log(exposure)) + maintenance_freq +
    supply_chain_index + energy_backup_score + solar_system,
    data = freq_bus_clean, init.theta = 0.1108406848, link = log)

Coefficients:
                              Estimate Std. Error z value Pr(>|z|)
(Intercept)                  -1.601490   0.049120 -32.603  < 2e-16 ***
maintenance_freq             -0.014851   0.007019  -2.116  0.03435 *
supply_chain_index            0.103412   0.048675   2.125  0.03362 *
energy_backup_score           0.030837   0.009934   3.104  0.00191 **
solar_systemHelionis Cluster -0.059332   0.038687  -1.534  0.12511
solar_systemZeta             -0.020798   0.031339  -0.664  0.50692
---
```

## Winning Severity Model: Gamma

```
Call:  glm(formula = claim.amount ~ base_salary + psych_stress_index +
    hours_per_week + protective_gear_quality + safety_training_index +
    experience_yrs + accident_history_flag + supervision_level +
    gravity_level + employment_type + solar_system, family = Gamma(link = "log"),
    data = work_sev_data2)

Coefficients:
                 (Intercept)                  base_salary           psych_stress_index
                   8.799e+00                    2.333e-05                    1.564e-01
              hours_per_week      protective_gear_quality        safety_training_index
                   1.455e-02                   -6.874e-02                   -9.951e-02
              experience_yrs        accident_history_flag             supervision_level
                  -9.444e-03                    1.919e-01                   -5.656e-01
               gravity_level     employment_typeFull-time  solar_systemHelionis Cluster
                  -1.411e+00                    1.752e-01                    6.065e-01
           solar_systemZeta
                   8.275e-01

Degrees of Freedom: 1825 Total (i.e. Null);  1813 Residual
Null Deviance:      4244
Residual Deviance: 3872         AIC: 35730
```

## Cargo Type B

## Winning Frequency Model: Negative Binomial

```
Call:
glm.nb(formula = claim_count ~ route_risk + cargo_type + container_type +
    weight + offset(log(exposure)), data = freq_cargo_typeB_clean,
    init.theta = 0.1313707587, link = log)

Coefficients:
                                      Estimate Std. Error z value Pr(>|z|)
(Intercept)                          -5.861e-01  7.542e-02  -7.771 7.78e-15 ***
route_risk                            7.497e-02  1.030e-02   7.277 3.41e-13 ***
cargo_typelithium                     7.650e-02  3.839e-02   1.993 0.046281 *
cargo_typerare earths                 8.268e-02  5.703e-02   1.450 0.147103
cargo_typesupplies                   -1.471e-01  4.682e-02  -3.142 0.001676 **
cargo_typetitanium                   -5.980e-02  3.861e-02  -1.549 0.121381
container_typeDockArc Freight Case    7.732e-03  3.582e-02   0.216 0.829119
container_typeHardSeal Transit Crate  1.206e-01  3.559e-02   3.387 0.000706 ***
container_typeLongHaul Vault Canister 8.102e-03  3.594e-02   0.225 0.821626
container_typeQuantumCrate Module     7.157e-02  3.576e-02   2.001 0.045342 *
weight                                8.535e-07  3.976e-07   2.146 0.031849 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for Negative Binomial(0.1314) family taken to be 1)

    Null deviance: 38741  on 83375  degrees of freedom
Residual deviance: 38641  on 83365  degrees of freedom
AIC: 113293

Number of Fisher Scoring iterations: 1


              Theta:  0.13137
          Std. Err.:  0.00192

 2 x log-likelihood:  -113269.23600
```

## Winning Severity Model: Gamma

```
Call:  glm(formula = claim_amount ~ cargo_type + weight + route_risk +
    solar_radiation + debris_density, family = Gamma(link = "log"),
    data = sev_cargo_typeB_clean)

Coefficients:
          (Intercept)      cargo_typelithium  cargo_typerare earths     cargo_typesupplies
            1.175e+01              4.620e-01              -1.295e-01              -1.571e+00
   cargo_typetitanium                 weight             route_risk        solar_radiation
           -1.890e+00              6.414e-06              1.518e-01              4.364e-01
       debris_density
            4.081e-01

Degrees of Freedom: 18610 Total (i.e. Null);  18602 Residual
  (38 observations deleted due to missingness)
Null Deviance:      24590
Residual Deviance: 11340        AIC: 504300
```

## Simulation Formula

Aggregate loss under 1 simulation: $\sum_{i=1}^{n} \sum_{j=1}^{N_i} X_{ij}$

Where:

- n: denotes the number of policies
- $N_i$: denotes the number of claims from policyholder 'i' (Frequency distribution)
- $X_{ij}$: denotes the size of the 'jth' policy from policyholder 'i' (Severity distribution)

# Appendix C: Pricing Results and Formulas

## Appendix C1: Scenario Testing Additional Results

| Workers Compensation | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Year | 2176 | 2177 | 2178 | 2179 | 2180 | 2181 | 2182 | 2183 | 2184 | 2185 | 2186 |
| Aggregate Loss (Scenario 1) | 3,156,698 | 4,399,936.63 | 4,632,536.64 | 4,887,375.26 | 5,148,240.87 | 5,424,608.74 | 5,717,475.77 | 6,024,401.30 | 6,348,418.91 | 6,690,512.35 | 7,050,356.20 |
| Profit Margin for scenario 1 | 13.03% | 12.07% | 12.44% | 13.11% | 13.84% | 14.57% | 15.37% | 16.20% | 17.02% | 17.84% | 18.65% |
| Aggregate Loss (Scenario 2) | 4,181,856.97 | 4,399,936.63 | 4,632,536.64 | 4,887,375.26 | 5,148,240.87 | 5,424,608.74 | 5,717,475.77 | 6,024,401.30 | 6,348,418.91 | 6,690,512.35 | 7,050,356.20 |
| Profit Margin for base case | 3.67% | 7.42% | 9.42% | 10.92% | 12.15% | 13.21% | 14.25% | 15.25% | 16.21% | 17.14% | 18.03% |
| Aggregate Loss (Scenario 3) | 5,008,028.00 | 4,399,936.63 | 4,632,536.64 | 4,887,375.26 | 5,148,240.87 | 5,424,608.74 | 5,717,475.77 | 6,024,401.30 | 6,348,418.91 | 6,690,512.35 | 7,050,356.20 |
| Profit Margin for scenario 3 | -3.88% | 3.67% | 6.99% | 9.16% | 10.79% | 12.11% | 13.34% | 14.49% | 15.56% | 16.57% | 17.54% |
| Aggregate Loss (Scenario 4) | 7,116,321.00 | 4,399,936.63 | 4,632,536.64 | 4,887,375.26 | 5,148,240.87 | 5,424,608.74 | 5,717,475.77 | 6,024,401.30 | 6,348,418.91 | 6,690,512.35 | 7,050,356.20 |
| Profit Margin for scenario 4 | -23.13% | -5.90% | 0.78% | 4.65% | 7.31% | 9.31% | 11.03% | 12.54% | 13.89% | 15.12% | 16.27% |
| Aggregate Loss (Scenario 5) | 8,868,728.00 | 4,399,936.63 | 4,632,536.64 | 4,887,375.26 | 5,148,240.87 | 5,424,608.74 | 5,717,475.77 | 6,024,401.30 | 6,348,418.91 | 6,690,512.35 | 7,050,356.20 |
| Profit Margin for scenario 5 | -39.13% | -13.85% | -4.38% | 0.91% | 4.42% | 6.99% | 9.11% | 10.92% | 12.50% | 13.92% | 15.22% |

| Equipment | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Year | 2176 | 2177 | 2178 | 2179 | 2180 | 2181 | 2182 | 2183 | 2184 | 2185 | 2186 |
| Aggregate Loss (Scenario 1) | 38,265,347 | 53,332,945.96 | 56,152,360.15 | 59,241,335.17 | 62,403,365.13 | 65,753,302.58 | 69,303,231.33 | 73,023,567.40 | 76,951,081.55 | 81,097,698.31 | 85,459,473.14 |
| Profit Margin for scenario 1 | 13.42% | 11.84% | 11.99% | 12.59% | 13.24% | 13.93% | 14.72% | 15.53% | 16.34% | 17.15% | 17.96% |
| Aggregate Loss (Scenario 2) | 50,689,537.28 | 53,332,945.96 | 56,152,360.15 | 59,241,335.17 | 62,403,365.13 | 65,753,302.58 | 69,303,231.33 | 73,023,567.40 | 76,951,081.55 | 81,097,698.31 | 85,459,473.14 |
| Profit Margin for base case | 3.78% | 7.07% | 8.90% | 10.33% | 11.51% | 12.54% | 13.57% | 14.58% | 15.51% | 16.43% | 17.33% |
| Aggregate Loss (Scenario 3) | 71,473,251.00 | 53,332,945.96 | 56,152,360.15 | 59,241,335.17 | 62,403,365.13 | 65,753,302.58 | 69,303,231.33 | 73,023,567.40 | 76,951,081.55 | 81,097,698.31 | 85,459,473.14 |
| Profit Margin for scenario 3 | -12.35% | -0.91% | 3.73% | 6.58% | 8.62% | 10.22% | 11.65% | 12.94% | 14.13% | 15.23% | 16.28% |
| Aggregate Loss (Scenario 4) | 90,389,501.00 | 53,332,945.96 | 56,152,360.15 | 59,241,335.17 | 62,403,365.13 | 65,753,302.58 | 69,303,231.33 | 73,023,567.40 | 76,951,081.55 | 81,097,698.31 | 85,459,473.14 |
| Profit Margin for scenario 4 | -27.02% | -8.18% | -0.97% | 3.17% | 5.99% | 8.10% | 9.90% | 11.47% | 12.87% | 14.14% | 15.32% |
| Aggregate Loss (Scenario 5) | 107,457,419.00 | 53,332,945.96 | 56,152,360.15 | 59,241,335.17 | 62,403,365.13 | 65,753,302.58 | 69,303,231.33 | 73,023,567.40 | 76,951,081.55 | 81,097,698.31 | 85,459,473.14 |
| Profit Margin for scenario 5 | -40.27% | -14.73% | -5.21% | 0.10% | 3.61% | 6.19% | 8.32% | 10.14% | 11.73% | 13.16% | 14.46% |

| Cargo Type B | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Year | 2176 | 2177 | 2178 | 2179 | 2180 | 2181 | 2182 | 2183 | 2184 | 2185 | 2186 |
| Aggregate Loss (Scenario 1) | 6,507,638,129.00 | 9,067,616,700.92 | 9,546,970,817.25 | 10,072,155,410.08 | 10,609,760,733.96 | 11,179,313,909.68 | 11,782,869,416.62 | 12,415,397,412.65 | 13,083,149,630.17 | 13,788,153,464.40 | 14,529,738,267.96 |
| Profit Margin for scenario 1 | 13.29% | 11.50% | 11.59% | 12.13% | 12.79% | 13.47% | 14.25% | 15.06% | 15.88% | 16.69% | 17.49% |
| Aggregate Loss (Scenario 2) | 8,618,186,873.65 | 9,067,616,700.92 | 9,546,970,817.25 | 10,072,155,410.08 | 10,609,760,733.96 | 11,179,313,909.68 | 11,782,869,416.62 | 12,415,397,412.65 | 13,083,149,630.17 | 13,788,153,464.40 | 14,529,738,267.96 |
| Profit Margin for base case | 3.51% | 6.65% | 8.46% | 9.86% | 11.04% | 12.07% | 13.09% | 14.08% | 15.04% | 15.96% | 16.86% |
| Aggregate Loss (Scenario 3) | 12,191,243,709.00 | 9,067,616,700.92 | 9,546,970,817.25 | 10,072,155,410.08 | 10,609,760,733.96 | 11,179,313,909.68 | 11,782,869,416.62 | 12,415,397,412.65 | 13,083,149,630.17 | 13,788,153,464.40 | 14,529,738,267.96 |
| Profit Margin for scenario 3 | -13.05% | -1.52% | 3.16% | 6.02% | 8.08% | 9.68% | 11.12% | 12.42% | 13.62% | 14.73% | 15.78% |
| Aggregate Loss (Scenario 4) | 18,286,865,564.00 | 9,067,616,700.92 | 9,546,970,817.25 | 10,072,155,410.08 | 10,609,760,733.96 | 11,179,313,909.68 | 11,782,869,416.62 | 12,415,397,412.65 | 13,083,149,630.17 | 13,788,153,464.40 | 14,529,738,267.96 |
| Profit Margin for scenario 4 | -41.31% | -15.49% | -5.88% | -0.53% | 3.02% | 5.62% | 7.76% | 9.59% | 11.19% | 12.63% | 13.94% |

| Bussiness Interruption | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Year | 2176 | 2177 | 2178 | 2179 | 2180 | 2181 | 2182 | 2183 | 2184 | 2185 | 2186 |
| Aggregate Loss (Scenario 1) | 37,670,947,059 | 52,489,840,493.63 | 55,264,684,417.42 | 58,304,827,866.03 | 61,416,871,375.31 | 64,713,851,864.48 | 68,207,662,127.25 | 71,869,185,845.57 | 75,734,612,510.92 | 79,815,678,134.53 | 84,108,500,530.45 |
| Profit Margin for scenario 1 | 13.40% | 11.80% | 11.72% | 12.27% | 12.93% | 13.61% | 14.38% | 15.18% | 15.98% | 16.78% | 17.57% |
| Aggregate Loss (Scenario 2) | 49,888,219,723.28 | 52,489,840,493.63 | 55,264,684,417.42 | 58,304,827,866.03 | 61,416,871,375.31 | 64,713,851,864.48 | 68,207,662,127.25 | 71,869,185,845.57 | 75,734,612,510.92 | 79,815,678,134.53 | 84,108,500,530.45 |
| Profit Margin for base case | 3.76% | 7.03% | 8.86% | 10.28% | 11.47% | 12.50% | 13.52% | 14.51% | 15.47% | 16.39% | 17.28% |
| Aggregate Loss (Scenario 3) | 59,455,598,728.00 | 52,489,840,493.63 | 55,264,684,417.42 | 58,304,827,866.03 | 61,416,871,375.31 | 64,713,851,864.48 | 68,207,662,127.25 | 71,869,185,845.57 | 75,734,612,510.92 | 79,815,678,134.53 | 84,108,500,530.45 |
| Profit Margin for scenario 3 | -3.80% | 3.29% | 6.44% | 8.53% | 10.11% | 11.41% | 12.62% | 13.76% | 14.82% | 15.83% | 16.79% |
| Aggregate Loss (Scenario 4) | 81,749,099,770.00 | 52,489,840,493.63 | 55,264,684,417.42 | 58,304,827,866.03 | 61,416,871,375.31 | 64,713,851,864.48 | 68,207,662,127.25 | 71,869,185,845.57 | 75,734,612,510.92 | 79,815,678,134.53 | 84,108,500,530.45 |
| Profit Margin for scenario 4 | -21.40% | -5.42% | 0.80% | 4.44% | 6.95% | 8.87% | 10.53% | 11.99% | 13.31% | 14.52% | 15.64% |
| Aggregate Loss (Scenario 5) | 105,934,088,378.00 | 52,489,840,493.63 | 55,264,684,417.42 | 58,304,827,866.03 | 61,416,871,375.31 | 64,713,851,864.48 | 68,207,662,127.25 | 71,869,185,845.57 | 75,734,612,510.92 | 79,815,678,134.53 | 84,108,500,530.45 |
| Profit Margin for scenario 5 | -40.50% | -14.87% | -5.32% | 0.00% | 3.53% | 6.12% | 8.25% | 10.07% | 11.67% | 13.09% | 14.40% |

## Appendix C2: Pricing Formulae

$$Pure\ Premium = \frac{Projected\ Aggregate\ Loss}{Exposure\ Units}$$

$$Gross\ Premium = \frac{Pure\ Premium}{1 - \%\ Risk\ Loading - \%\ Costs}$$

$$Profit\ Margin\ (without\ reserves) = \frac{Total\ Profit}{Total\ Revenue}$$

$$Reserves\ in\ Year\ 2176 = 0.5 \cdot Profit_{2175} + 0.05 \cdot (Total\ Premium)_{2176}$$

**For all years after 2176 (i.e. years 2177 - 2186)**

$$Reserves_t = Reserves_{t-1} + 0.05 \cdot (Total\ Premium)_t$$

5% of Total Premiums collected in each year will be set aside as reserves. Such reserves will accumulate from year-to-year given their recursive calculation method.

$$Investment\ Income\ in\ Year\ 2176\ (with\ reserves)$$
$$= (0.5 \cdot Profit + 0.95 \cdot Total\ Premium_{2176}) \cdot (1 + interest\ rate_{2176})$$

**For all years after 2176 (i.e. years 2177 - 2186)**

$$(Investment\ Income\ with\ reserves)_t$$
$$= ((Investment\ Income\ with\ reserves)_{t-1} + 0.95 \cdot Total\ Premium_t) \cdot (1 + interest\ rate_t)$$

95% of Total Premiums collected in each year will be set aside as investment income. Investment income (with reserves) will accumulate from year-to-year according to the current interest rate.

$$Total\ Premium = Single\ Premium \cdot Total\ Units$$

$$Profit\ Margin\ (with\ reserves)$$
$$= \frac{(Investment\ Income\ with\ reserves)_t - (Investment\ Income\ with\ reserves)_{t-1} - Aggregate\ Loss}{Total\ Premium - (Investment\ Incomee\ with\ reserves)_t - (Investment\ Income\ with\ reserves)_{t-1}}$$

**END OF PRICING FORMULAE**

## Appendix D: Workers Classification

**White Collar:**

- Management: Executive, Vice President, Director
- Administration: HR, IT, Legal, Finance & Accounting
- Environmental & Safety: Environmental Scientists, Safety Officer, Medical Personnel

**Non-White Collar**

- Exploration Operations: Geoligist, Scientist, Field Technician Geologist, Scientist, Field technician
- Extraction Operations: Drilling Operators, Maintenance, Engineers, Freight Operators, Drilling operators, Maintenance, Engineers, Freight operators, Robotics technician
- Spacecraft Operations**:** Navigation Officers, Maintenance, Security Personnel, Steward, Galleyhand

## Reference List


Chief Risk Officer Forum & Allianz Commercial. (2025). *"Understanding and Mitigating the Impact of Space Related Risks".* Accessed by: https://commercial.allianz.com/content/dam/onemarketing/commercial/commercial/reports/commercial-cro-eri-space-risk.pdf

Lander & Rodgers. (April 4, 2024). *"Failure in orbit: Planning your Space Projects with Space Insurance in Mind"*. Accessed by: https://www.landers.com.au/legal-insights-news/failure-in-orbit-planning-your-space-projects-with-space-insurance-in-mind

Safe Work Australia. (2023). "*Comparative Performance Monitoring Report 23: Workers' Compensation Premiums" Accessed by:* https://www.safeworkaustralia.gov.au/sites/default/files/2021-11/CPM%2023%20-%20Premiums.pdf

Gallagher. (June 24, 2024). *"When the stars are the limit: Insuring a future space economy"*. Accessed by: https://specialty.ajg.com/plane-talking/insuring-a-future-space-economy


## AI Acknowledgements


- AI was used to identify and correct minor coding errors.
- AI was used to refine sentence structure and to ensure coherence in the report, thereby improving readability.


**END OF APPENDIX**